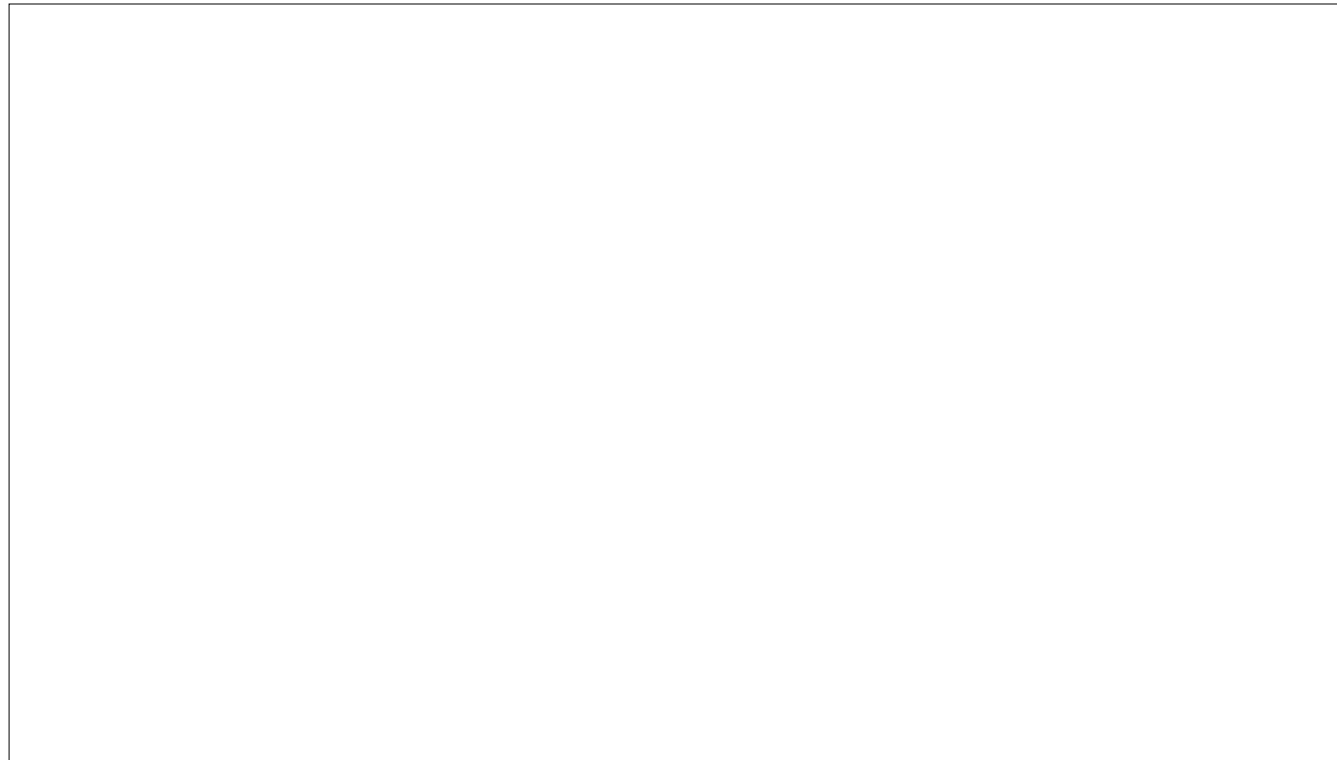




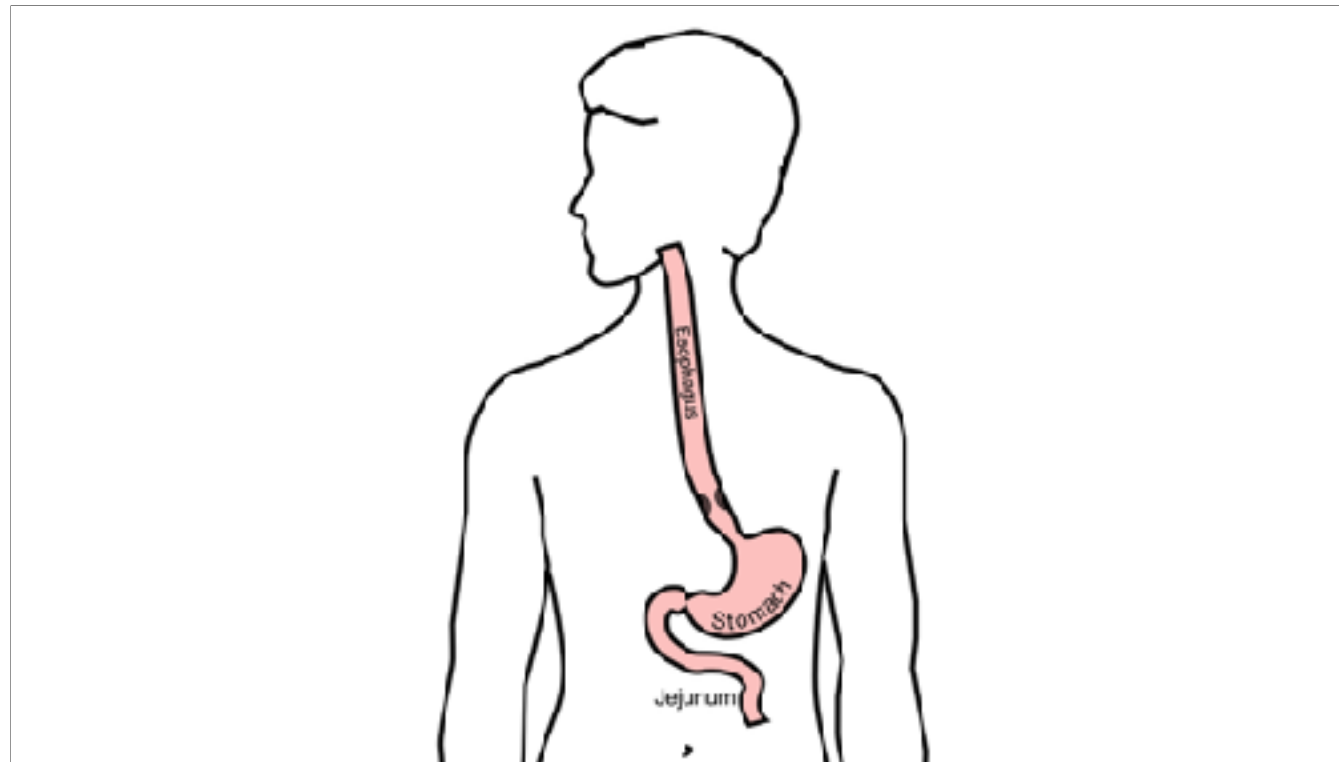
I'm Dr Jonathan Salo, a GI Cancer Surgeon in Charlotte, North Carolina.

These videos are designed to help you and your family make the right decisions about your cancer care.

While I am a surgeon, I'm not *your* surgeon. There is of course no substitute for the advice of the cancer care team that knows you.

Jejunostomy

The topic for this video is a feeding jejunostomy



We will start with a review of anatomy. Food travels from the esophagus to the stomach, and finally into the jejunum, the first part of the small intestine.

Esophagus



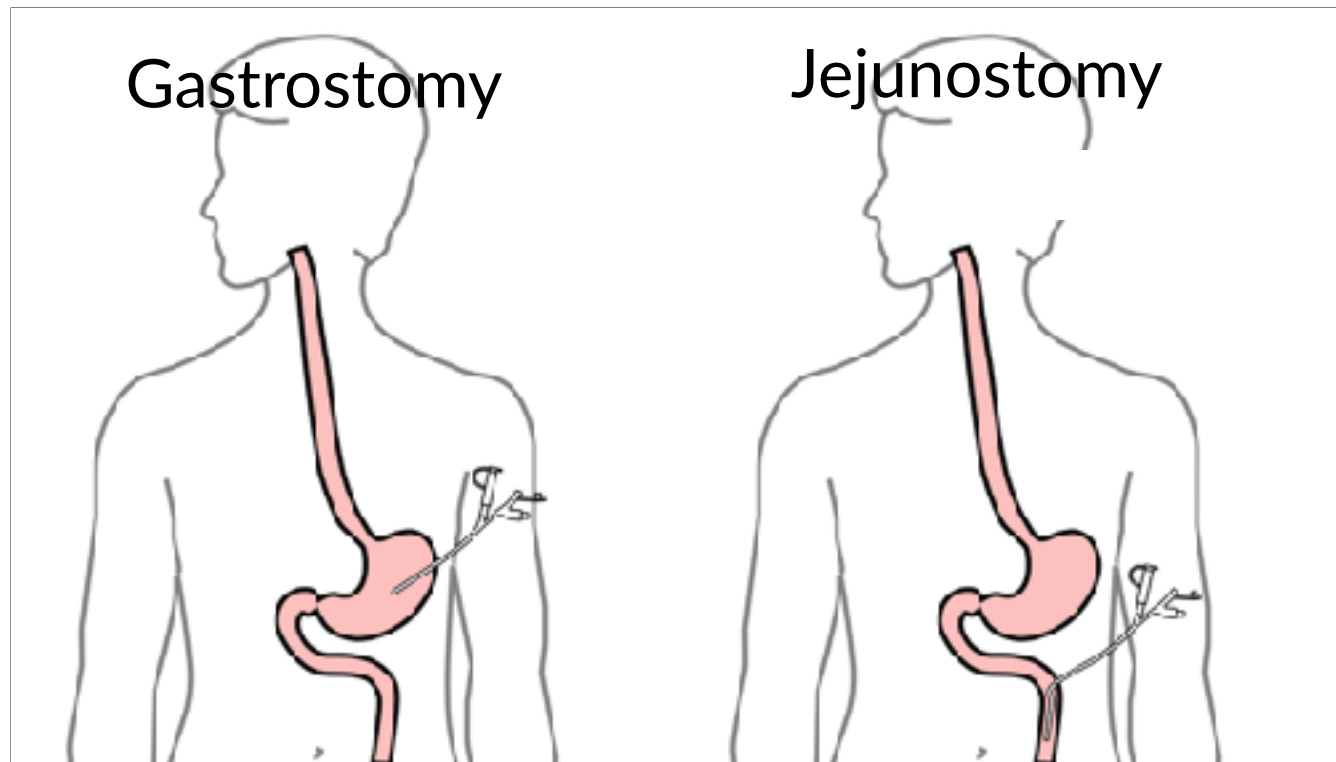
Stomach



Cancers in the esophagus or stomach tend to grow for a long period of time without symptoms.

With time, however, they can cause difficulty with eating and in extreme cases can cause weight loss

Some patients with cancer of the esophagus or stomach can benefit from a feeding tube to help with nutrition



There are two basic kinds of feeding tubes:

A gastrostomy tube is placed through the skin into the stomach

A jejunostomy tube is placed through the skin into the small intestine

Jejunostomy



This video focuses on the feeding jejunostomy

Jejunostomy Tube

- Formula
- Water
- Medicines

A jejunostomy tube can be used to administer

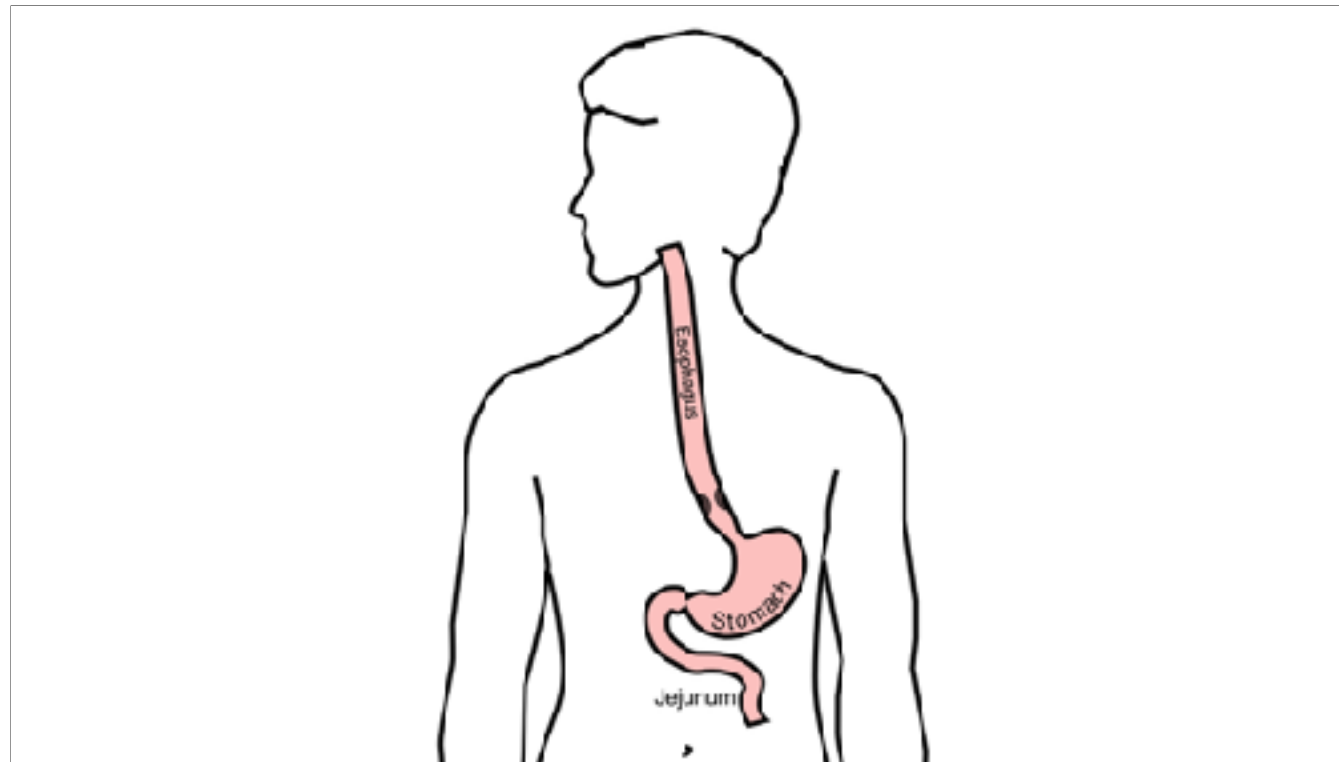
Formula

Water

And medicines



Here are some examples of the formula which might be used with your feeding tube. It is specially designed to provide protein, carbohydrates, fats, and vitamins and minerals



If we go back to our drawing, we see that the stomach is quite large, which allows it to store food after a meal.



This makes it possible to have a large Thanksgiving meal, and spend the next few days digesting it.



Stomach

The stomach acts as a temporary storage area for food, like this grain elevator



The stomach sends a small amount of food at a time into the jejunum, much like this grain elevator sends grain onto this conveyor belt.

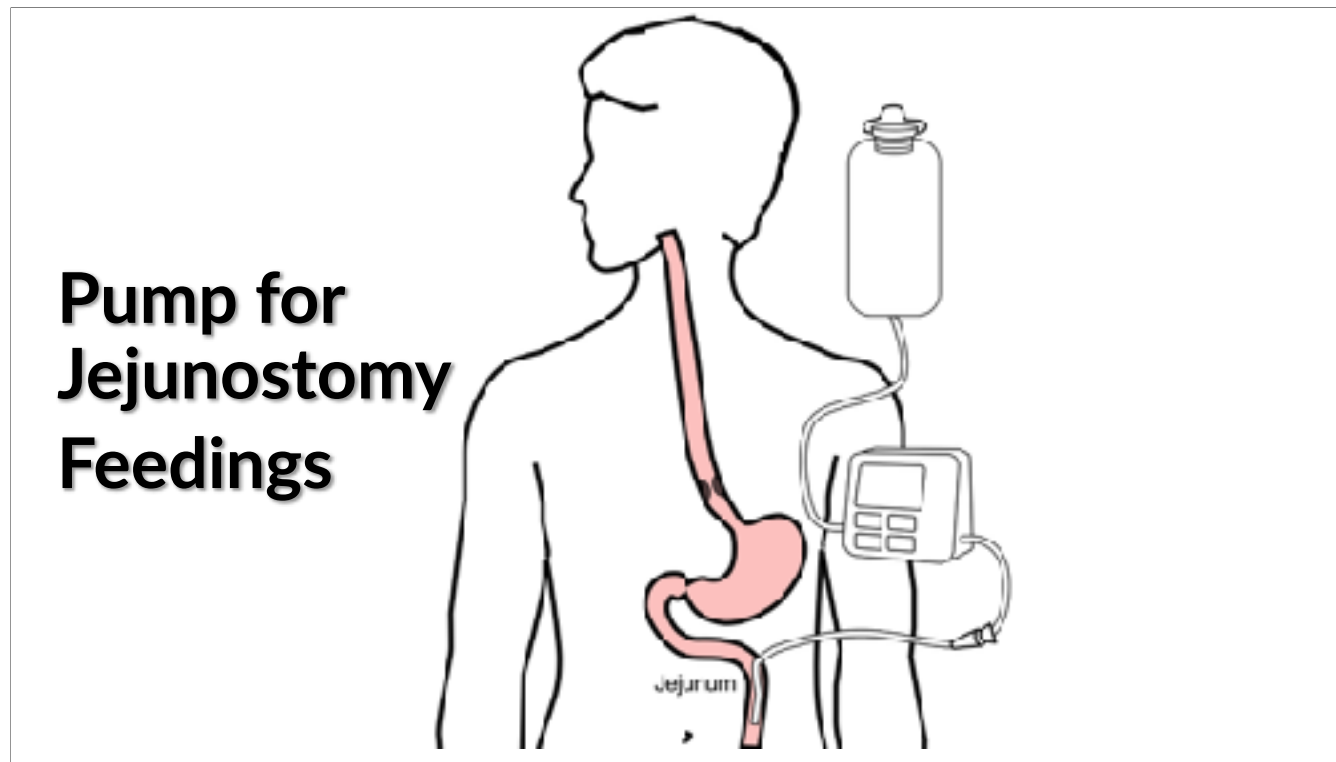


Just like a conveyor belt, it's easy for the jejunum to get overloaded with food.



Jejunum
= Small Intestine

... and if it's overloaded, it will start to complain.



So when you have a feeding jejunostomy, it's important that the formula goes into the tube as a slow, steady rate in order to avoid overloading the jejunum. This requires a pump to administer the feeding over a several hours.



Here's an example of a typical pump and feeding bag used with a jejunostomy tube



And here's the bag and a pump connected to the feeding tube.

You can move around the house with a bag and pump, but its best to run the feedings at night, so that you be more mobile during the day.

How Much Formula Do I Need?

An important question is “How much formula do I need?”



The short answer is that your dietitian will calculate how much formula you need, but we can give you a rough idea.

Rate and Duration

5 cartons: $240\text{mL}/\text{carton} \times 5 = 1200\text{mL}$
○ $50\text{mL}/\text{hour} \times 24 \text{ hours} = 1200\text{mL}$

Your dietitian will calculate how many cartons of tube feeds you will need

For instance, if that's 5 cartons and each carton is 240mL, the total will be 1200mL per day

One method would be to run the tube feeds continuously, 24 hours per day.

This would require a rate of 50mL per hour, because 50×24 is 1200.

5 cartons Tube feeds 1200mL

24 hours/day: 50mL/hour

So we could run tube feeds 24 hours per day at 50mL/hour. This would work fine...

5 cartons Tube feeds

24 hours/day: 50mL/hour→Inconvenient

The problem with this is it's inconvenient.

If you want to go for a walk or go shopping, you would need to take your pump with you,



If you can avoid it, you don't want to feel like you're chained to your infusion pump.

5 cartons Tube feeds

24 hours/day: 50mL/hour → Inconvenient

16 hours/day: 75mL/hour

An alternative would be to increase the rate and decrease the time.

You could increase the rate to 75mL/hour, which would only take 16 hours. You would have 8 hours a day disconnected from the pump.

5 cartons Tube feeds

24 hours/day: 50mL/hour→Inconvenient

16 hours/day: 75mL/hour

12 hours/day: 100mL/hour

So what if you wanted to get the feeding in within 12 hours? You would need to run the pump at 100mL/hour.

For many patients, this won't work, because the small intestine can't handle that much formula at a time



Too high a rate of tube feeds will overwhelm the jejunum, and the result is cramps and diarrhea.

5 cartons Tube feeds

24 hours/day: 50mL/hour→Inconvenient

16 hours/day: 75mL/hour

12 hours/day: 100mL/hour→Diarrhea/cramps

A slow rate of 50mL/hour is inconvenient, as it may require being hooked up to the pump all day.

On the other hand, a faster rate of 100mL/hr would be more convenient, but carries the risk of diarrhea and cramps.

5 cartons Tube feeds

16 hours/day: 75mL/hour

Running tube feeds for 16 hours a day is for many patients a good compromise between convenience and good tolerance of tube feeds. Most patients can tolerate 75mL/hour of tube feeds.

Typical Tube Feedings 150lbs+

$$5 \text{ cartons} \times 240\text{mL}/\text{carton} = 1200\text{mL}$$

People with a body weight over 150 pounds usually need about 5 cartons of feeds.

Each carton is about 240mL

So the total amount is 1200 mL

Typical Tube Feedings 150lbs+

5 cartons x 240mL/carton = 1200mL

75mL/hour x 16 hours = 1200mL

This is usually administered at a rate of 75mL/hour for 16 hours, which totals 1200mL

Typical Tube Feedings 150lbs+

5 cartons x 240mL/carton = 1200mL

75mL/hour x 16 hours = 1200mL

6pm to 10am

In the hospital we run the tube feeds from 6pm to 10am, but at home you can of course use whatever schedule works best for you.

Typical Tube Feedings <150lbs

$$4 \text{ cartons} \times 240\text{mL}/\text{carton} = 960\text{mL}$$

Those with a body weight *less* than 150 pounds generally need 4 cartons of feeds.

Each carton is 240mL, so the total is 960mL

Typical Tube Feedings <150lbs

4 cartons x 240mL/carton = 960mL

60mL/hour x 16 hours = 960mL

This can be administered at a rate of 60mL/hour over 16 hours

Typical Tube Feedings <150lbs

4 cartons x 240mL/carton = 960mL

60mL/hour x 16 hours = 960mL

6pm to 10am

Again this usually is from 6pm to 10am

Starting Tube Feeds

Feeds need to be started slowly

- Allows jejunum to adjust
- Typically start at 30mL/hour (continuous)

When starting tube feeds for the first time it is important to start slowly to allow the small intestine, the jejunum, to get adjusted.

We typically start at 30mL/hour and run them continuously, meaning 24 hours per day.

Starting Tube Feeds

Increase rate 10mL/hour every 8-12 hours

We would then increase 10mL/hour every 8-12 hours

Starting Tube Feeds

- Monday 7am - start at 30mL/hour

For instance, if you started tube feeds for the first time at 7am on a Monday morning...

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour

You might increase to 40mL/hour 8 hours later at 3pm

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour

And increased to 50mL/hour 8 more hours later at 11pm

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour
- Tuesday 7am - Increase to 60mL/hour

And then increase to 60mL/hour 8 hours later at 7 am the next day

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour
- Tuesday 7am - Increase to 60mL/hour
- Tuesday 3pm - Increase to 70mL/hour

... and increase to 70mL/ hour 8 hours later at 3pm.

Once your body has adjusted to the tube feeding, you can change the duration from continuous to just at night, or nocturnal

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour
- Tuesday 7am - Increase to 60mL/hour
- Tuesday 3pm - Increase to 70mL/hour
- Tuesday 6pm - Increase to 75mL/hour

... and increase to 70mL/ hour 8 hours later at 3pm.

Once your body has adjusted to the tube feeding, you can change the duration from continuous to just at night, or nocturnal

You might increase to 75mL/hour at 6pm, and then run the tube feeds for 16 hours, and stop at 10am the next day.

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour
- Tuesday 7am - Increase to 60mL/hour
- Tuesday 3pm - Increase to 70mL/hour
- Tuesday 6pm - Increase to 75mL/hour
- Wednesday 10am - Stop tube feeds (16 hours)

Wednesday morning at 10am, the feedings would have been running for 16 hours, and you can stop the feeding tube pump.

Starting Tube Feeds

- Monday 7am - start at 30mL/hour
- Monday 3pm - Increase to 40mL/hour
- Monday 11pm - Increase to 50mL/hour
- Tuesday 7am - Increase to 60mL/hour
- Tuesday 3pm - Increase to 70mL/hour
- Tuesday 6pm - Increase to 75mL/hour
- Wednesday 10am - Stop tube feeds (16 hours)
- Wednesday 6pm - Start tube feeds at 75mL/hour

Wednesday at 6pm, you would then start tube feeds again, to run until 10am the next day

Typical Tube Feed Plans

Nocturnal feeds: 16 hours/day (6p-10am)

- 5 cartons: 75mL/hour x 24 hours
- 4 cartons: 60mL/hour x 24 hours

So a typical plan tube feeds is 16 hours per day, which in our example is 6pm to 10am

For patients who need 5 cartons of feeds, getting the feeding done in 16 hours requires a rate of 75mL/hour.

For those needing 4 cartons, getting the feeding done in 16 hours requires a rate of 60mL/hour

Jejunostomy Tube

- Water
- Pump
- Syringe

The jejunostomy tube can also be used to administer water.

Water

Water is required to prevent dehydration
Can be taken by mouth or by feeding tube

In addition to formula, water is required to prevent dehydration.

Most patients need about a liter of water a day, which can be taken either by mouth or through the feeding tube

One way to do this is taking 240mL of water 4 times per day.

Water

Water is required to prevent dehydration
Can be taken by mouth or by feeding tube
 $240\text{mL} \times 4 \text{ times per day} = 960\text{mL} \sim 32\text{oz}$

In addition to formula, water is required to prevent dehydration.

Most patients need about a liter of water a day, which can be taken either by mouth or through the feeding tube

One way to do this is taking 240mL of water 4 times per day.

Water

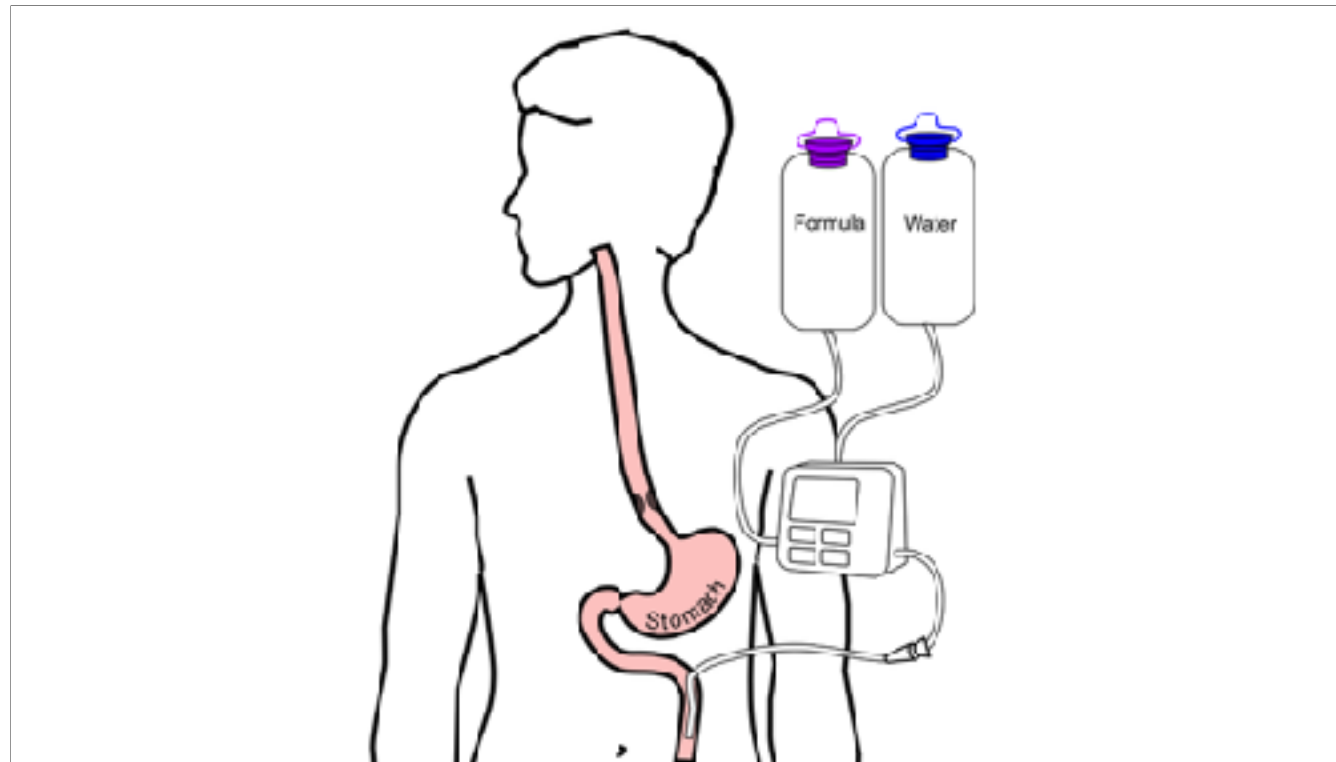
240mL x 4 times per day = 960mL

8oz x 4 times per day = 32oz

In addition to formula, water is required to prevent dehydration.

Most patients need about a liter of water a day, which can be taken either by mouth or through the feeding tube

One way to do this is taking 240mL of water 4 times per day.

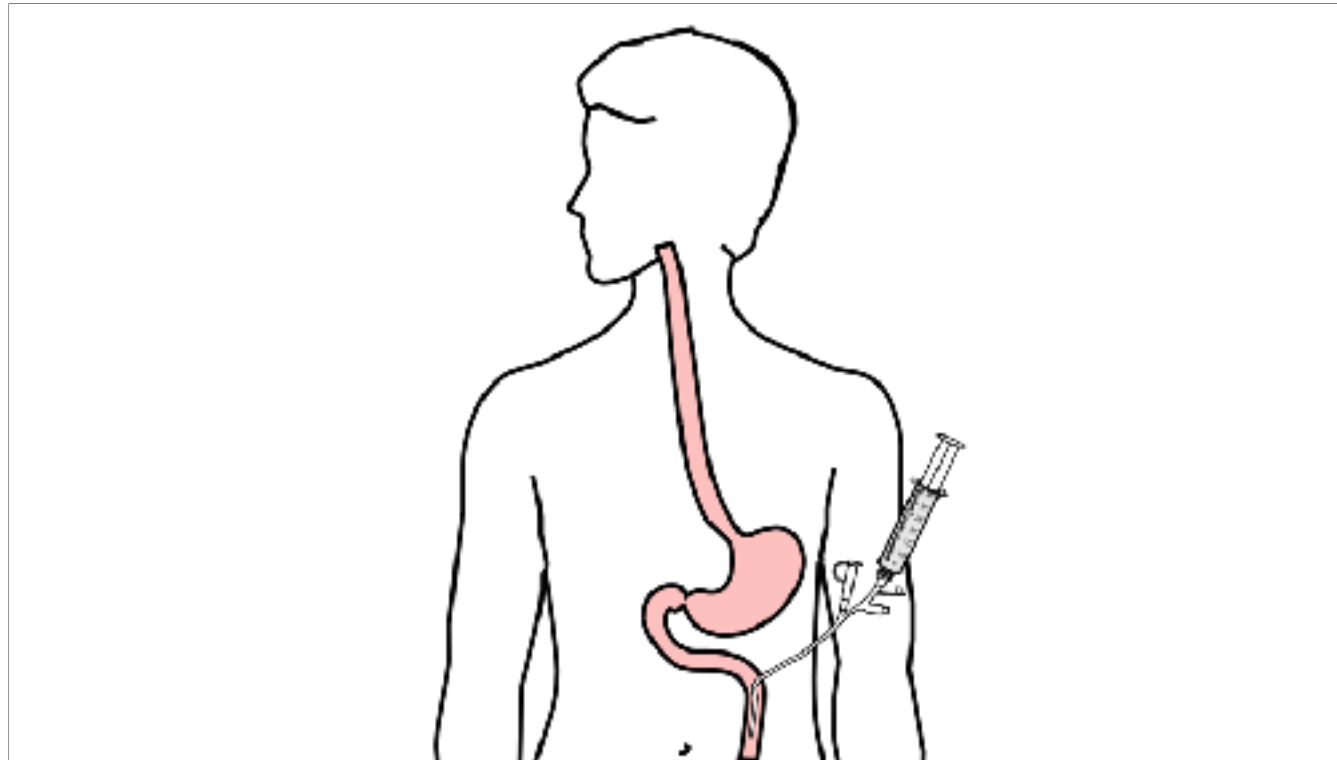


Some pumps come with two bags, one for formula and one for water.

Your care team will tell you how much water to use, but it's generally around one quart or 1000mL per day total



It's also possible to use a syringe to administer water



Water can be flushed into the jejunostomy tube slowly with a syringe.

While it won't work to administer a syringe full of *formula* into a jejunostomy tube, it is usually fine to inject a syringe full of *water*.

Jejunostomy Tube

- Formula
- Water
- Medicines

A feeding jejunostomy can also be used to administer medicines.

Administer Medicines

Liquid Medicines

A few medicines are available in liquid form



An example is the pain reliever Acetaminophen, also known by the brand name of Tylenol

Administer Medicines

Liquid Medicines

Crush pills and Resuspend in water

Some pills can be crushed and resuspended in water

Medicines which can't be crushed

Extended Release	XR	XL	XT
Sustained Release	ER	SR	LA
Controlled Release	CR	CD	

A word of caution, however, as some pills can't be crushed:

Extended release,
Sustained release, or
Controlled release

Medicines which can't be crushed

Extended Release
Sustained Release
Controlled Release

Tamsulosin (Flomax)
Liquid-filled capsules
Soft-gel capsules

In addition, Flomax, or tamsulosin, can't be given via a tube.

The same with liquid-filled capsules and soft-gel capsules.

Your pharmacist can help here

Medicine Administration

Crush pills to powder → Dissolve in 60ml water

Many medicines, however, can be crushed and resuspended in water.



You place a pill inside the chamber of a pill crusher



Screw the cap on and twist the cap back and forth to crush the tablet inside the chamber



You now have a fine powder

Medicine Administration

Crush pills to powder → Dissolve in 60ml water

...and you can dissolve the power in water, typically 60mL



The water is introduced into the chamber, and once the powder is dissolved,



the liquid is drawn up into the syringe.

Medicine Administration

Crush pills to powder → Dissolve in 60ml water
Inject medicine into feeding tube

The medicine can then be injected into the feeding tube

Medicine Administration

Crush pills to powder → Dissolve in water

Inject medicine into feeding tube

Flush tube with additional water

It is important to flush the tube with additional water after administering the medicine

Medicine Administration

Don't mix medicines together

It's important to administer the medicines separately, and not mix them together

Medicine Administration

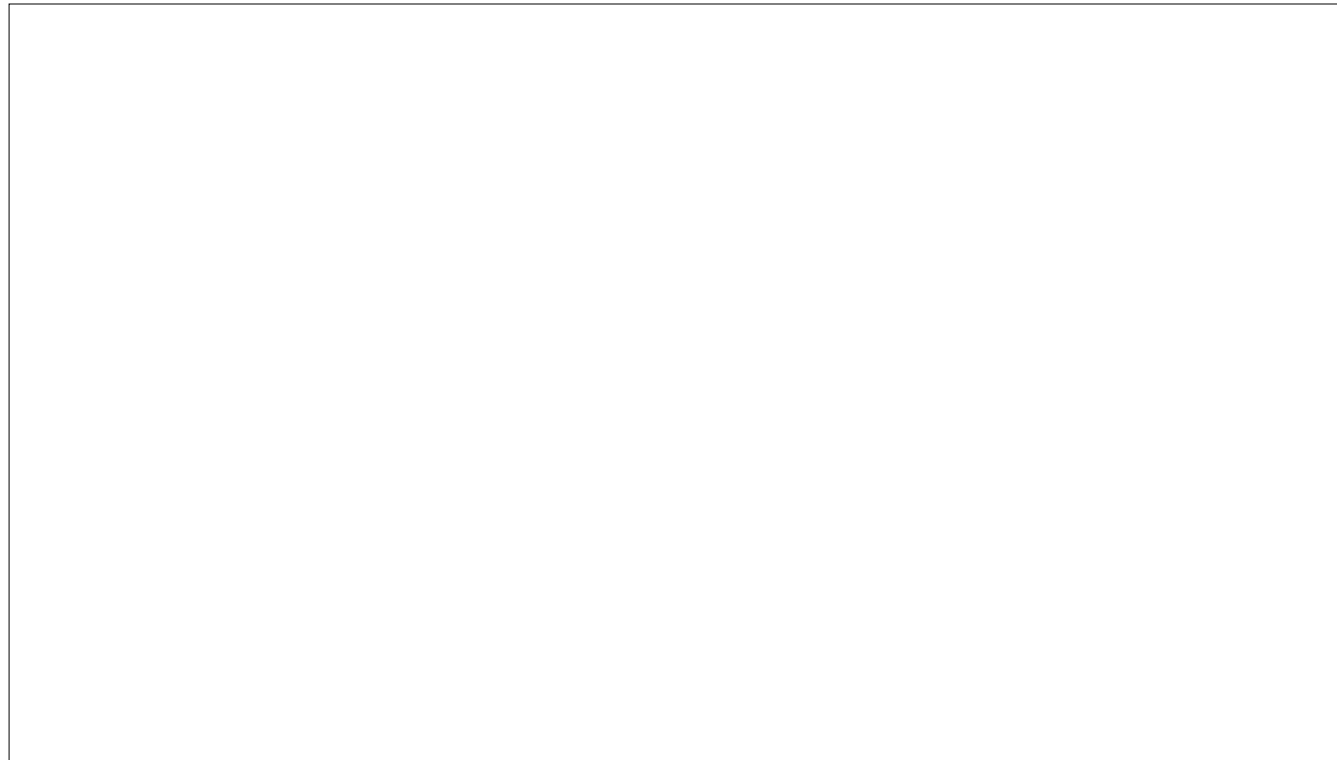
Don't mix medicines together
Flush tube between medicines

It is also important to flush the tube between medicines

Medicine Administration

Don't mix medicines together
Flush tube between medicines
Check with your pharmacist

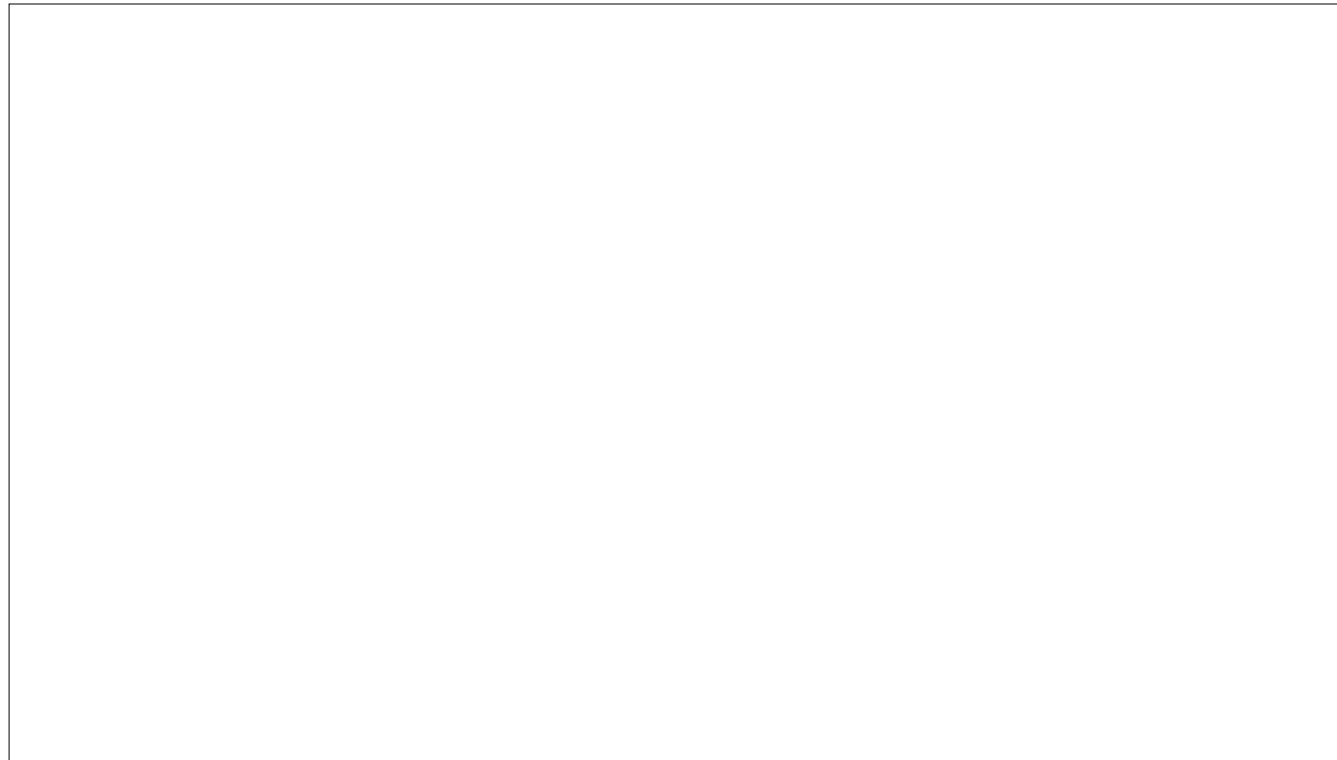
Your pharmacist can help by making certain that the medicines you are taking are suitable to be crushed and given through a feeding tube.



We hope you have found this video helpful. We have a companion video which describes a feeding gastrostomy.

Please leave a comment if you have topics you would like covered in videos.

Feel free to subscribe to be notified of new videos as we make them.



Our friend Brad has a feeding tube, and we will demonstrate how to administer water

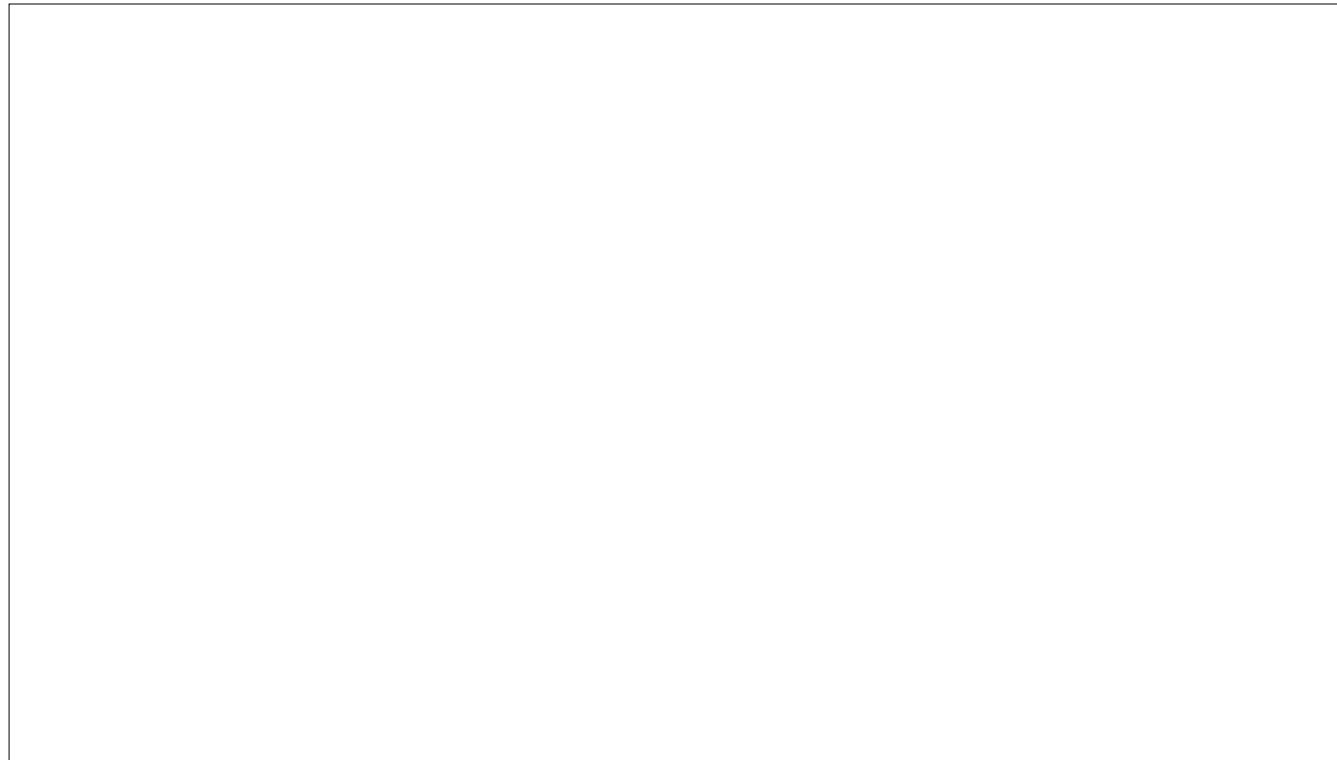
Open the cap

Attach the syringe to the tube by twisting it on

Empty the syringe

Remove the syringe by twisting

Close the cap



To use the funnel method, first remove the plunger from the barrel of the syringe

Open the cap

Attach the syringe to the tube by twisting it on

The water is simply poured into the syringe barrel and gravity will take it into the jejunum.

Remove the syringe by twisting

Close the cap